

Introducing the Cross-Chain Interoperability Protocol (CCIP) for Decentralized Inter-Chain Messaging and Token Movements

August 5, 2021 11 min read

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The growth of numerous independent [blockchain](#) ecosystems with varying optimizations and geographical niches has resulted in an increasingly multi-chain world. Being able to seamlessly tap into the advantages of each of these blockchains and their unique assets within a single application would drive a massive wave of new cross-chain smart contract development—no different than the proliferation of [DeFi](#), [NFT](#), and [on-chain gaming](#) economies when [decentralized oracle services](#) for real-world data and secure off-chain computation were introduced.

However, building cross-chain applications is notoriously difficult given the problems with existing cross-chain infrastructure. For one, there is extensive fragmentation in token bridges and messaging protocols, most being application-specific services between two distinct chains. Additionally, many bridges are fairly centralized with weak security guarantees, lack transparent or reliable node operators, and drive up costs and processing times for end-users. These limitations and vulnerabilities have already resulted in exploitation to the tune of tens of millions of dollars in funds lost, stunting cross-chain innovation.

In order to accommodate the growing ecosystem demand for inter-chain solutions, we're excited to announce the upcoming launch of the [Cross-Chain Interoperability Protocol \(CCIP\)](#)—a new open standard for cross-chain communication. CCIP aims to establish a universal connection between

hundreds of blockchain networks, both private and public, unlocking isolated tokens and empowering cross-chain applications for all on-chain ecosystems.



CCIP provides [smart contract](#) developers a generalized, compute-enabled infrastructure for transferring data and smart contract commands across blockchain networks. CCIP will underpin a variety of cross-chain services, such as the Chainlink Programmable Token Bridge, which will empower users to move their tokens across any blockchain network in a highly secure, scalable, and cost-efficient manner.

Watch Chainlink Co-founder Sergey Nazarov announce CCIP at the [Smart Contract Summit #1](#):

The Cross-Chain Interoperability Protocol is the culmination of years of research and development in secure cross-chain technology, dating back to the original intentions of the [Chainlink 1.0 Whitepaper](#). CCIP is being spearheaded by Chainlink Labs Chief Scientist Ari Juels, Chainlink Labs VP of Engineering Ben Chan—the architect behind the widely used cross-chain Wrapped Bitcoin token on Ethereum (WBTC)—and numerous world-class researchers working with the Chainlink Labs team.

In the following post, we outline why Chainlink is optimized to provide a generalized cross-chain communication protocol for the blockchain industry, how CCIP is incorporated within a multi-layered technology stack, and how

CCIP will power various services to support the development of cross-chain [hybrid smart contracts](#).

Why Chainlink Is Optimized for Generalized Cross-Chain Infrastructure

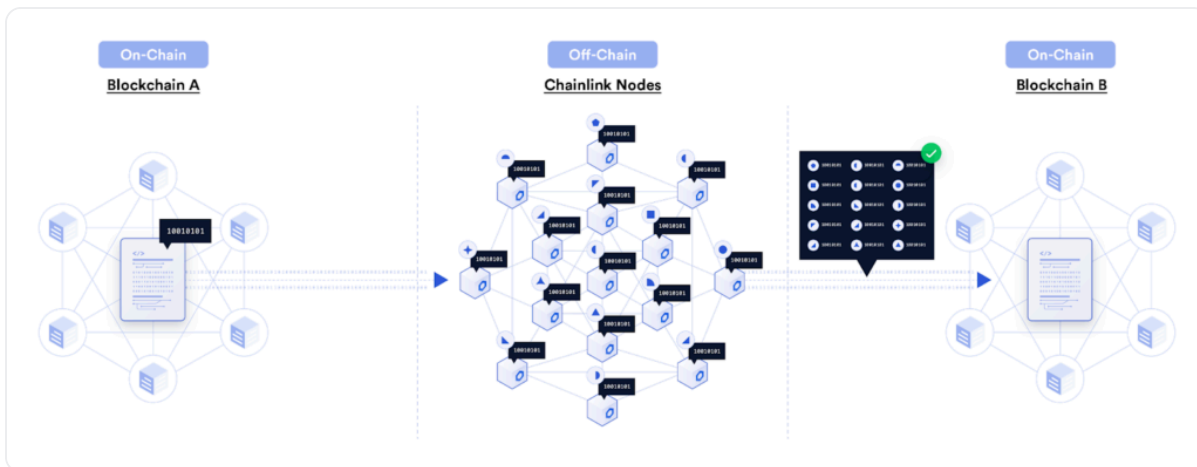
Building secure and efficient cross-chain technology for both token movements and generalized messaging across all blockchain networks is not a simple task. However, the historical reliability of Chainlink's existing infrastructure, alongside a growing Chainlink ecosystem and specific cross-chain technological innovations makes Chainlink the most well-suited protocol to be a global standard for cross-chain communication.

Decentralized Network of Proven Node Operators

[Cross-chain bridges](#) at their most basic level are a committee of nodes that collectively attest to information on one chain and relay it to another by cryptographically signing transactions in a threshold manner. The Chainlink Network is already supported by the largest collection of independent, Sybil-resistant, and provably reliable node operators in the industry, made up of some of the top DevOps and infrastructure providers in the world. Chainlink's growing network of node operators secures over \$30B+ for smart contract ecosystems through numerous oracle services and will further scale in size via a coming upgrade to Chainlink's [Off-Chain Reporting protocol](#) (OCR).

Chainlink OCR 1.0 is a secure and efficient off-chain computation protocol for data aggregation that has operated at scale for an extended period of time without fail, successfully lowering on-chain gas costs of oracle reports by up to 90%. OCR 2.0 will expand upon this success by enabling even more efficient and expressive off-chain computations that give rise to advanced cross-chain capabilities. CCIP will leverage OCR 2.0 within its protocol stack to scale the number of nodes that sign committee-based reports into the hundreds, leading to increased security of locked funds while maintaining a high degree of cost-efficiency for users. Combining the world's largest pool of

secure node operators with upgraded off-chain computation capabilities, CCIP will achieve both a high degree of tamper resistance and performance.



Chainlink OCR 2.0 aggregates node responses off-chain and commits the oracle report on-chain as a single transaction.

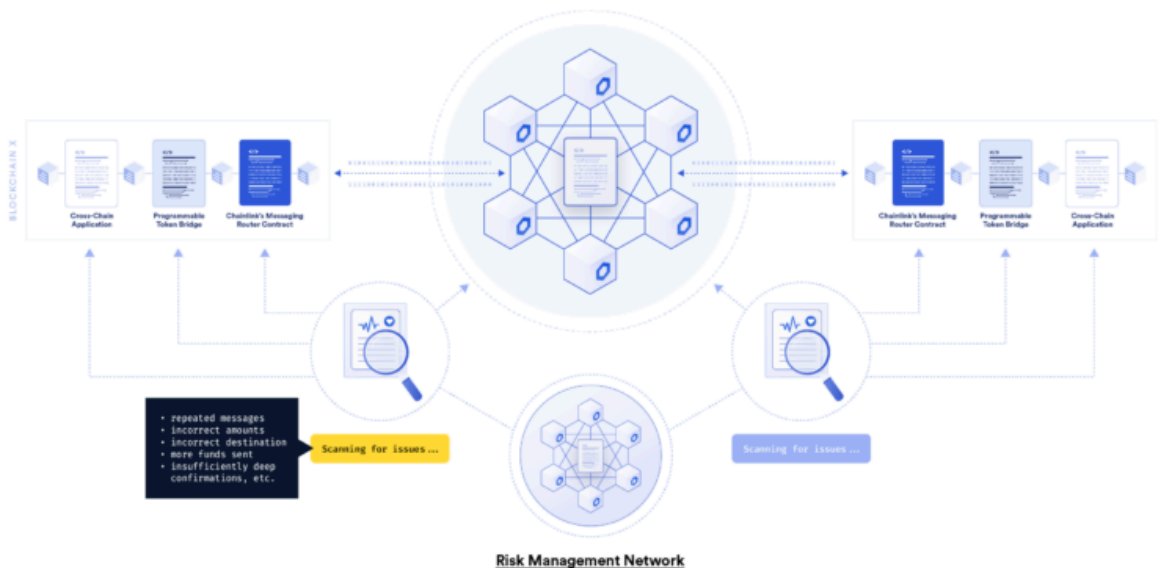
Risk Management Network

Security is a foundational element for cross-chain services aiming to directly secure high-value contracts. To that end, CCIP is secured by a newly-invented risk management system never before seen in the blockchain industry called the Risk Management Network. The Risk Management Network will consist of decentralized oracle networks with the sole purpose of monitoring CCIP services for malicious activity that could lead to financial loss. Importantly, the Risk Management Network will contain fully independent committees of nodes compared to the node committees they are responsible for monitoring on CCIP, completely separating anomaly detection and cross-chain services.

The Risk Management Network acts as a verification layer and will periodically submit heartbeat checks when the system is operating as normal. If the Risk Management Network's heartbeat messages stop or its nodes notice any nefarious activity, an emergency shutdown is automatically triggered to stop a particular cross-chain service. The pause allows user funds to be protected against a potential black swan event affecting the service. While the Risk Management Network will initially consist of high-quality Chainlink nodes independent of the CCIP services they protect, dApps securing a large amount of value through a CCIP service can join the Risk

Management Network to provide their users with greater guarantees that any anomalous activity will be caught and mitigated.

The Risk Management Network is a decentralized implementation of risk management commonly used when securing high-value contracts. The Risk Management Network revolutionizes how risk is managed in cross-chain infrastructure by establishing a checks and balances system that separates responsibilities and minimizes the control any single group has over the operation of CCIP services. Importantly, the network can also scale and evolve into the future through additions like AI for enhanced detection techniques.



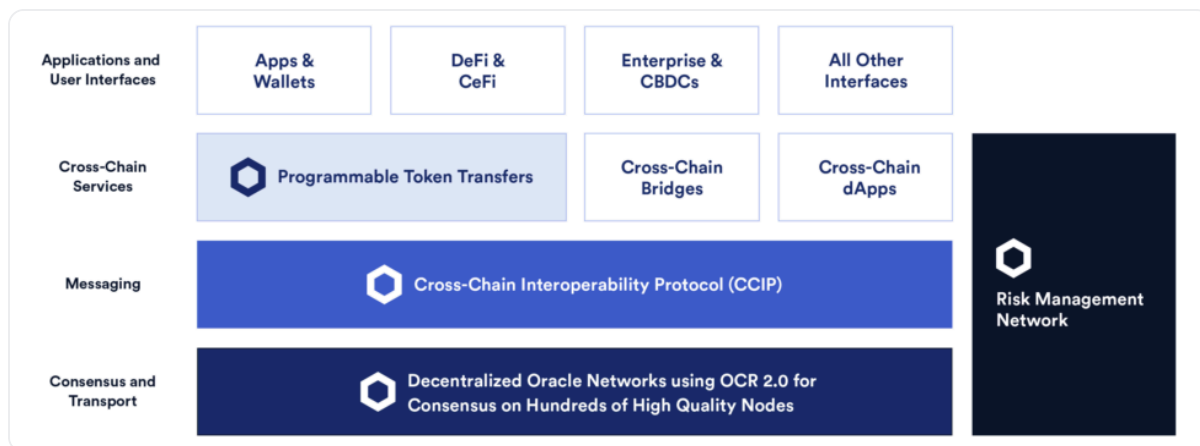
Large Ecosystem Support Across Blockchains

Cross-chain systems are most useful when they have large network effects. Network effects lead to increased security of user funds, increased token access and a simplified user experience, better documentation and tooling for developers, and greater revenue opportunities for token provisioning no matter which blockchain an asset is natively launched on. With over a hundred blockchains working with Chainlink and many blockchains and layer-2 solutions already integrated with Chainlink on mainnet, Chainlink is the ideal infrastructure to serve as a credibly neutral protocol for cross-chain communication between all blockchains.

On top of being the most widely integrated [oracle](#) solution by top blockchains, Chainlink powers a large number of dApps running across these blockchains. Chainlink already works with the top lending, insurance, and other DeFi protocols and is accelerating adoption within supported chains through open-source development and the [Chainlink Community Grant Program](#). Numerous blockchain partners and applications have already expressed a desire to use Chainlink oracles for cross-chain activities, making it a top priority to bring the entire smart contract industry a highly secure, reliable, and performant set of generalized cross-chain solutions.

Defining the Cross-Chain Tech Stack

The Cross-Chain Interoperability Protocol (CCIP) sits within a layered technology stack that will be leveraged to provide users with new cross-chain services, such as the Chainlink Programmable Token Bridge, various other bridge implementations, and the ability to create powerful cross-chain applications that span any blockchain network. Each layer of the tech stack provides a key role in ultimately supporting the expanding multi-chain ecosystem.



The Cross-Chain Interoperability Protocol (CCIP) is part of a layered tech stack for supporting data and asset movements across blockchains.

User Interfaces

At the top of the technology stack are the interfaces that enable users to connect to the Chainlink Programmable Token Bridge or other bridge implementations and begin moving their tokens across blockchain

environments. Importantly, new and existing ecosystem projects can deploy their own interface in a permissionless manner. This can include established wallets, aggregators, applications, exchanges, and a wide range of user-facing services wanting to provide a gateway to the multi-chain ecosystem. By taking a community-driven approach to the interface layer, cross-chain infrastructure can become easily accessible, censorship-resistant, and innovative in terms of how users interact with solutions built upon the CCIP.

Programmable Token Bridge

The Programmable Token Bridge is a reference bridge implementation built upon the CCIP that allows developers to build cross-chain applications that seamlessly and securely move their existing tokens across any blockchain. It's a unified bridge system, where various bridge connections between chains are secured by unique committees of nodes in order to distribute security while maintaining universal interoperability using routing contracts. The Programmable Token Bridge will support existing token standards, meaning already liquid assets can be instantly used within different smart contract ecosystems. On top of employing high-quality node operators and decentralized architecture, additional safety precautions will be implemented, such as time-based flow limits to minimize downside risk during black swan events—the parameters of which can be managed by much larger DONs.

The Programmable Token Bridge is compute-enabled, empowering users and smart contracts to send not only their tokens but also commands to the bridge and have it run customized logic around how it interacts with other blockchains. Users don't need to know how to use other blockchains—they just need to send instructions to the bridge on how they want to interact with other chains, and the bridge will automatically move the tokens cross-chain and deploy them in smart contracts on the destination chain within an atomic transaction. Thus, a user can stay on their blockchain of choice while still benefiting from smart contract ecosystems on other blockchain networks. The Programmable Token Bridge opens up advanced hybrid smart contract use cases like cross-chain yield aggregators, collateralized loans, and much more. Importantly, the Programmable Token Bridge is just one reference implementation built by Chainlink Labs, but any third-party bridge application

can be easily built by independent development teams wanting to take advantage of the security and functionality of CCIP.

Cross-Chain Interoperability Protocol (CCIP)

CCIP is a global standard that enables smart contracts on any blockchain to send and receive data packets to and from smart contracts on any other blockchain network. The protocol is generalized in nature, supporting the delivery of any type of data that a smart contract may want to deliver across chains. Generalized cross-chain interoperability will give developers a simple framework to build cross-chain applications without dealing with the complexity of the underlying protocols.

All applications built on top of CCIP like the Programmable Token Bridge, other cross-chain bridges, and cross-chain dApps can tap into the security of the Risk Management Network in order to automate detection and mitigation of nefarious activity. We are also exploring defense-in-depth approaches that can be deployed in the future such as trusted execution environments, hardware security modules for key management, crypto-economic security through staking, and more.

Network Infrastructure

Chainlink's cross-chain tech stack will be powered by decentralized networks of independent, highly reputable oracle node operators. Chainlink nodes will run the OCR 2.0 client to securely and cost-efficiently reach consensus off-chain on cross-chain transactions. OCR 2.0 removes any single point of failure and supports the ability to scale to hundreds of independent node operators without meaningfully increasing on-chain gas costs. Importantly, the report will contain a signature of every oracle node that responded, creating accountability and providing a trail that can be used to implement trust-minimization techniques.

Cross-Chain Applications Seamlessly Connected to All Blockchains

The introduction of CCIP is designed to rapidly expand what developers can build on each blockchain. Completely novel cross-chain applications become possible that leverage tokens on all chains simultaneously as well as take advantage of the unique properties of specific chains or the applications and assets on those chains.

This opens up a whole new crop of cross-chain applications like cross-chain protocols that leverage the scalable computation of one blockchain, the token diversity of another, the storage of a third, and the settlement security of a fourth to create a single hybrid smart contract application with superior functionalities. There are truly limitless possibilities and we are excited to bring an initial implementation to the blockchain industry in the near future.

Whether you're a user, interface, application, or existing cross-chain bridge, we are interested in hearing your feedback on CCIP for its continued improvement. Please share any feedback at support@chain.link. Through community feedback, CCIP can accommodate the cross-chain needs of all blockchain ecosystems and uses cases, ultimately driving an immense wave of development and innovation that takes the hybrid smart contract industry across all ecosystems to the next level of adoption and maturation.

If you're interested in building cross-chain functionality with CCIP and want to learn more, visit <https://chain.link/cross-chain> to learn more and [reach out to an expert](#).

To learn more about Chainlink, visit chain.link, subscribe to the [Chainlink newsletter](#), and follow [@chainlink](#) on Twitter.

ANNOUNCEMENTS

CCIP

CROSS-CHAIN